

SERVICE MANUAL

IGC-Medical Advances Inc.
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53226

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Revision 0

GE Signa[®] EXCITE[™] 3.0T Quadrature Extremity Coil

GE Catalog Part Number: M1385AM

Supplier Part Number: 14280

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All packages should be closely examined at time of delivery. If damage is apparent, have notation "**damage in shipment**" written on **all** copies of the freight or express bill **before** delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage **MUST** be reported to the carrier **immediately** upon discovery, or in any event, within **14** days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this **14** day period.

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- Call 1-800-548-3366 and use option 8.
http://data.supportcentral.ge.com/upload/15325/doc_324531.doc
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Rev. 08/15/2003

Language Policy For Service Documentation (Dir. 2128126)

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- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓLO EXISTE EN INGLÉS
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
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ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTADO REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTES AVISOS PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

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SECTION 1 - INTRODUCTION

1-1 Product Identification and Shipping List



SHIPPING LIST – TABLE 1-1

Description	GE Part#	Supplier Part#	Qty
GE Signa EXCITE 3.0T Quadrature Extremity Coil	2380635-2	14280	1
Knee Pad	2357819-3	14629	1
Leg/Foot Support Pad	2357819-4	14627	1
Foot Support Pad	2357819-5	14626	1
Toe Pad	2357819-6	14628	1
Phantom Assembly	2357819-12	14246	1
Phantom Positioner	2357819-11	14270	1
Operator's Manual	2380635-3	2380635-3	1
Service Manual	2380635-4	2380635-4	1
Material Safety Data Sheet	2357819-13	14249	1

1-2 Compatibility

This coil is compatible with the following hardware configuration:

- GE Signa EXCITE 3.0T G3 Systems

1-3 Related Documentation

Signa EXCITE 3.0T Service Methods CD-ROM, 2389160-1

GE Medical Systems Signa EXCITE 3.0T Quadrature Extremity Coil Operator Manual, 2380635-3

1-4 Environmental Requirements

Storage Requirements

Store the coil in an air-conditioned scan room or equipment room.

To store the coil, a storage space of (55cm) x (45cm) x (40cm) is required.

Dimensions

19" Wide X 12" Deep X 12" High (48.26 cm Wide X 30.48 cm Deep X 35.56 cm High)

Weight

15 lbs (6.81 Kg)

1-5 Theory of Operation

The Quadrature Lower Extremity Coil is a twelve (12) rod transmit-receive birdcage design. In MR systems, a birdcage coil is a near field antenna, which generates a homogeneous transmit RF magnetic field within its volume and, based on the reciprocity principle, has the same homogeneity as an RF receiver. The transmit field can be either linearly or circularly polarized, and is for this device, circularly polarized.

The coil consists of 2 distinct parts: The system independent antenna (which is comprised of the Bottom Coil and the Top Coil) and the cable section (which contains the Combiner and Bazooka Balun). Refer to Appendix 7-2.

During the transmit phase, RF power is provided to the transmit port (TX) of the system connector and the power is passed to the hybrid combiner. The power is split into two signals of equal amplitude and 90 degree phase difference (Mode 1 and Mode 2). These signals pass through the Bazooka Balun to remove any standing waves. The signals are then coupled to the antenna via a match and tune circuit and a balun circuit and generate a circularly polarized magnetic field useable for MRI. During the receive or detection phase, the MRI signal is received by the antenna and coupled to the system cable via the balun circuit and the match and tune circuit passing through the Bazooka Balun to the Combiner and then routed to the receive port (RX) of the system connector.

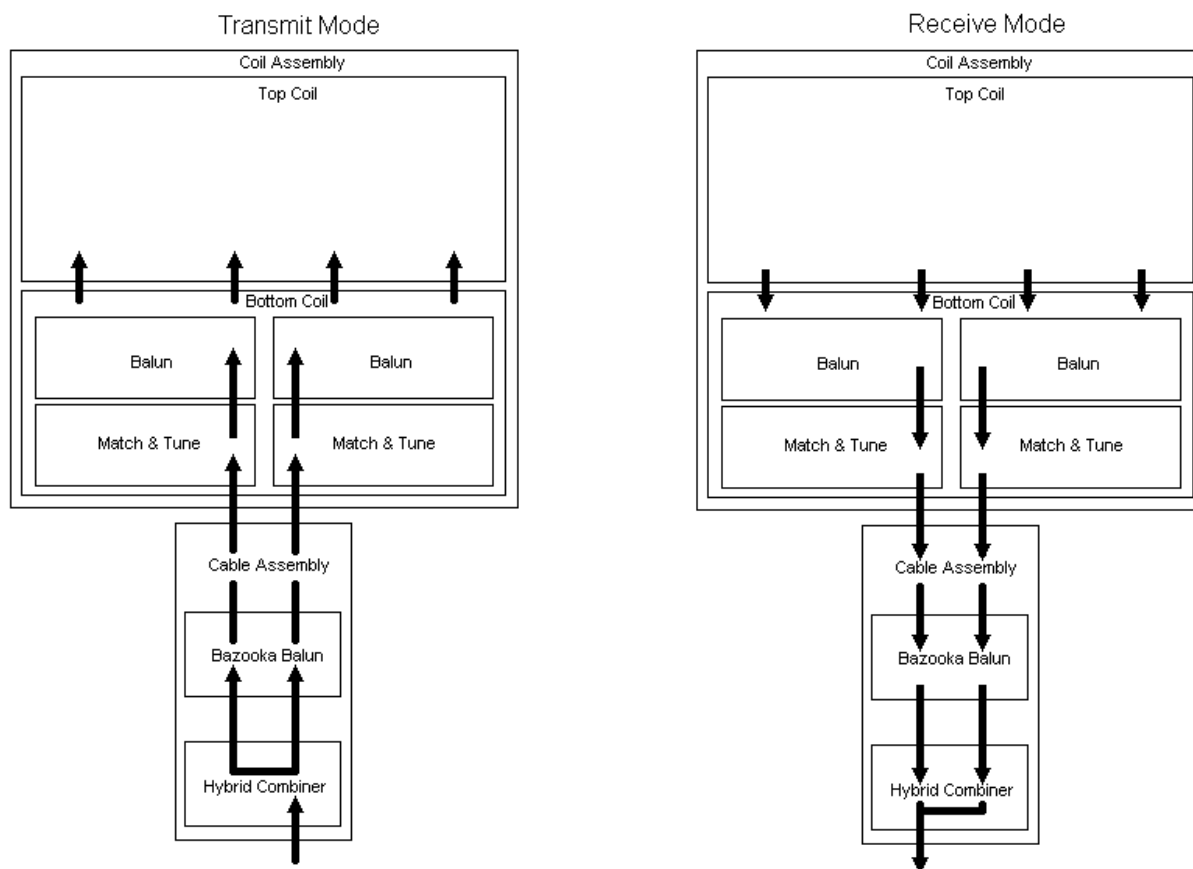


Figure 1-5: Signa EXCITE 3.0T Extremity Coil Block Diagram

SECTION 2 - SETUP AND CALIBRATION

2-1 Coil Installation

2-1-1 Special Install Notes

Check that Coil Polarity matches System Polarity. Coils are shipped for forward ramped magnets. For reverse ramped magnets, see comments in Section 5-2.

2-1-2 Installing the Coil

The name for this coil is: **Quadknee**

Add the coil using the Configuration File Manager. Refer to: Service Methods CD; Software; System Level Software Utilities; Configuration File Manager.

If the coil does not exist in the Coil Config File refer to the Adding New Coils to Config File Manager procedure and use the coil configuration information in Section 7-3 of this manual.

2-2 Installation Functional Checks

1. Perform system level Signal to Noise Check. Refer to Service Methods CD; Functional Checks; System Level Functional Checks; Signal to Noise Check.
2. Perform Section 3-2 - Coil Imaging Performance Verification.

2-3 Periodic Quality Assurance Check

On a periodic basis, such as during planned maintenance, perform the quality assurance checks as outlined below to ensure the coils is operating properly.

1. Check external cable for cracks or cuts.
2. Perform Section 3-2 - Coil Imaging Performance Verification and record data values in Data Sheet.

SECTION 3 - FUNCTIONAL CHECKS

3-1 Scanner Verification

Perform system level Signal to Noise Check. Refer to Service Methods CD; System Level Procedures; Functional Checks; Signal to Noise Check.

3-2 Coil Imaging Performance Verification

3-2-1 Tools Required

TOOLS REQUIRED – TABLE 3-2-1

Description	GE Part#	Supplier Part#	Qty
Signa EXCITE 3.0T Extremity Coil Phantom	2357819-12	14246	1
Signa EXCITE 3.0T Extremity Coil Phantom Positioner	2357819-11	14270	1

3-2-2 Explanation of Procedure

The Signa EXCITE 3.0T extremity coil can be used in one (1) mode of operation and has one (1) coil name: **QUADKNEE**. Refer to the Data Sheet in Appendix 7-1 to understand the data required to calculate the individual element SNR.

The image quality check uses two different protocols for signal and noise image acquisition. The signal scan is an **FSE** sequence used to minimize susceptibility and B₀ inhomogeneity effects. The noise scan is a **GRE** sequence that has a Control Variable (do_noise) to eliminate the transmit RF completely during the scan. The signal scan **must** be run prior to the noise scan as the R1, R2, and TG values from the signal scan are used for the noise scan.

3-2-3 Signal Scan

1. From the Scan Desktop, start new scan by selecting [**New Pt**]; set **Patient ID** to “geservice” and **Patient Weight** to “111” pounds. Click [**Patient Position**] to open protocols window.
2. Remove all other surface coils from the cradle. Place the coil on the cradle as illustrated in Figure 3-2-3-1. The coil should be positioned such that its cables exit towards the magnet. Remove any pads from the coil. Set the phantom positioner into the coil as seen in Figure 3-2-3-1.

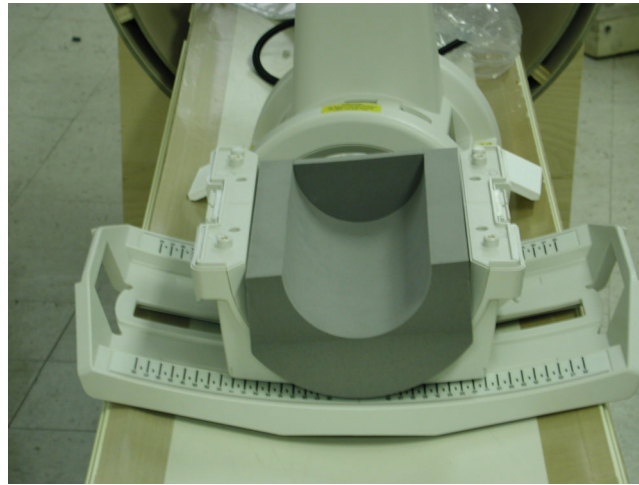


Figure 3-2-3-1: Placement of posterior section and phantom positioner

Center the coil within the baseplate and place the phantom in the phantom positioner as shown in Figure 3-2-3-2.

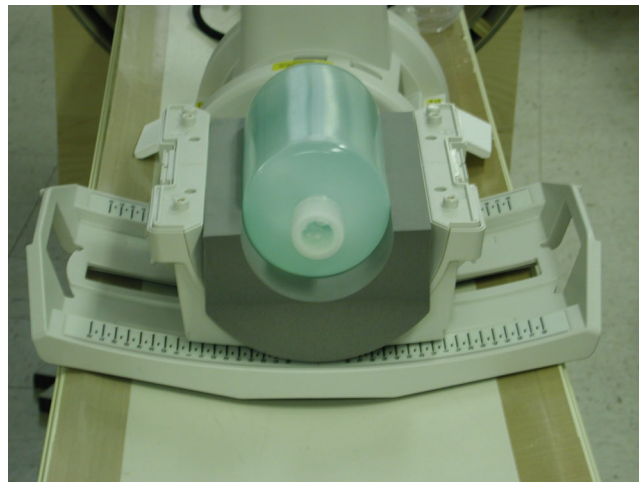


Figure 3-2-3-2: Phantom position

Attach the anterior portion of the coil.

3. At the magnet, press “**Alignment Light**” button to turn on the light. Move the cradle to align the coil to the alignment lights as shown in Figure 3-2-3-3. Note the alignment markings on the coil in front of the ‘chimney’. Press “**Landmark**” button to landmark the alignment.

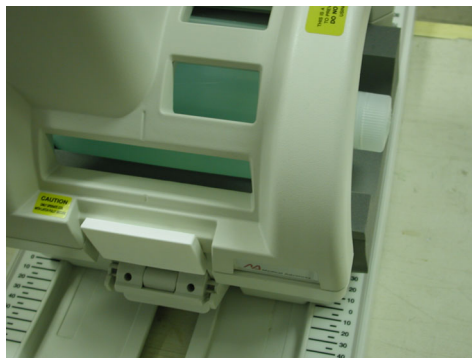


Figure 3-2-3-3: Coil alignment

4. Move the coil to scan position by pushing the “**Move to Scan**” button, ensuring cable does not get snagged.
5. At the console, set the protocols per the Signal section from Table 3-2-4: Signal and Noise Protocols. GE Service Personnel can use the Extremity SNR Signal protocol in the GE, Other protocols section.
6. Click [**Save Series**] to download the protocols, then click [**Prepare to Scan**].
7. Run [**Auto Prescan**]. Take note of the R1, R2 and TG values.
8. Run [**Scan**]

3-2-4 Noise Scan

A signal scan must be run prior to the noise scan as the same R1, R2 and TG values must be used for both the signal and noise scans. Do not run an Auto Prescan prior to the noise scan as the values will be changed.

1. Copy the signal scan series. Use [**Copy Series**] (highlight signal series and click right mouse button) and [**Paste Series**] in **RX Manager**.
2. Click [**View Edit**] and set the protocols per the Noise section from Table 3-2-4: Signal and Noise Protocols. GE Service Personnel can use the Extremity Coil SNR Noise protocol in the GE, Other protocols section.
3. Click [**Save Series**] and click [**Prepare to Scan**].
4. Open [**Display CVs**] menu under [**Research Operations**]. Set the “rhformat” and “do_noise” CVs to “1”.
5. Run [**Manual Prescan**], and do not make any changes. Verify that R1, R2, TG and Bandwidth have not changed from the previously completed signal scan, and click [**Done**].
6. Run [**Scan**].

SIGNAL AND NOISE PROTOCOLS – TABLE 3-2-4

Protocol	Signal	Noise
Protocol	Signal	Noise
<i>Patient Exam Information</i>		
Patient ID	geservice	geservice
Patient Name		
Patient Weight	111 lbs. (50 kg)	111 lbs. (50 kg)
<i>Patient Position</i>		
Patient Position	Supine	Supine
Patient Entry	Head First	Head First
Coil	QUADKNEE	QUADKNEE
Series Description	Signal	Noise
<i>Imaging Parameters</i>		
Plane	Axial	Axial
Mode	2D	2D
Pulse Seq	FSE-XL	GRE
Imaging Options	Fast (or None)	None
PSD Name	<i>Leave Blank</i>	<i>Leave Blank</i>
Protocol	<i>Leave Blank</i>	<i>Leave Blank</i>
<i>Scan Timing</i>		
# of Echoes	1	1
TE	17.0	Min Full
TR	500.0	34
Echo Train Length	4	N/A
Bandwidth	15.63	15.63
<i>Additional Parameters</i>		
Flip Angle	N/A	1
<i>Acquisition Timing</i>		
Freq	256	256
Phase	256	256
NEX	1.00	1.00
Phase FOV	1.00	1.00
Freq DIR	R/L	R/L
Auto Center Freq	Peak	Peak
Autoshim	On	On
Phase Correct	On	N/A
Contrast	Off	Off
# of Reps B4 Pause	0	N/A
<i>Scanning Range</i>		
FOV	20	20
Slice Thickness	3.0	3.0
Spacing	1.5	1.5
S/I Start	0.0	0.0
R/L Center	0.0	0.0
A/P Center	0.0	0.0
S/I End	0.0	0.0
# Slices	1	1
Table Delta	0.00	0.00

3-2-5 SNR Image Analysis

SNR Measurement

A circular ROI should be used for the signal measurements, and a rectangular ROI should be used for the noise measurements. The actual ROIs should be similar in shape and size to those in Figure 3-2-5 below.

ROIs in both signal and noise images can be measured directly in the image browser. Click the user interface button [MEASURE], select the circular or rectangular shape, and adjust its size and orientation so that the ROI is similar to those found in Figure 3-2-5. Mean, standard deviation, and area of the ROI will appear in the lower right corner of the image.

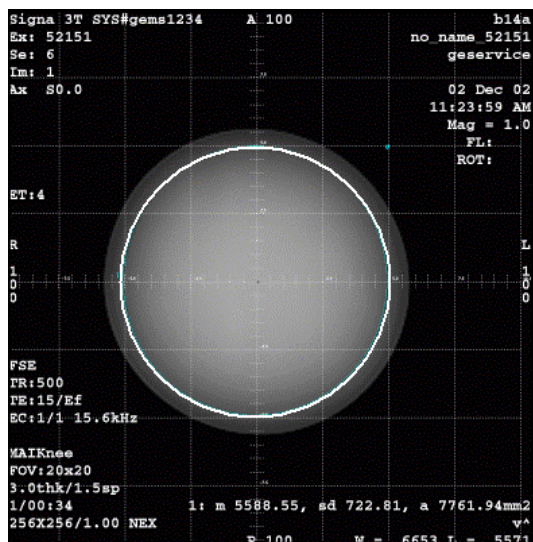
For the signal measurement, choose a 10 cm diameter circular ROI. This should be approximately a 5600 mm². For the noise measurement, choose a 19 cm rectangular ROI. This should be approximately 34000 mm².

Record the values for mean signal, standard deviation of noise, and SNR in Table 7-1.

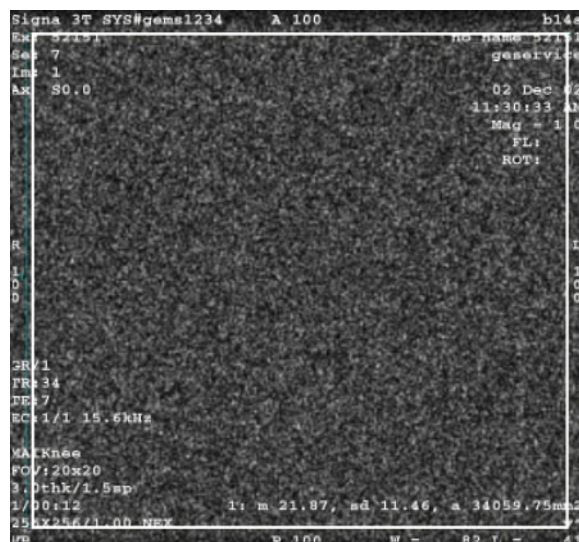
Individual receiver SNR is defined as the mean of data within the signal ROI divided by the standard deviation of data within the noise ROI.

$$SNR = \frac{\text{Mean of Signal within ROI}}{\text{Std.Deviation of Noise within ROI}}$$

Note: The SNR calculation uses the **MEAN** of the signal image and **STANDARD DEVIATION** of the noise image.



A: Signal ROI Measurement (10cm dia.)



B: Noise ROI Measurement (19cm)

Figure 3-2-5: Example ROIs

SNR Specification

The SNR measurements must be greater than or equal to the following specifications:

SNR SPECIFICATIONS – TABLE 3-2-5

Plane (if applicable)	SNR
Signa EXCITE 3.0T Quad Extrem Coil	410

3-3 External Cable Check

1. Remove four (4) screws from the Bottom Cover of the Interface Box Assembly.
2. Remove the Top Coil Assembly.
3. Remove four (4) large nylon screws from the Bottom Coil Assembly.
4. Carefully lift the Bottom Inner Assembly out of the Bottom Cover and set it down on the Cable Assembly.
5. Using the DMM, verify continuity of the cable by performing the following operations:

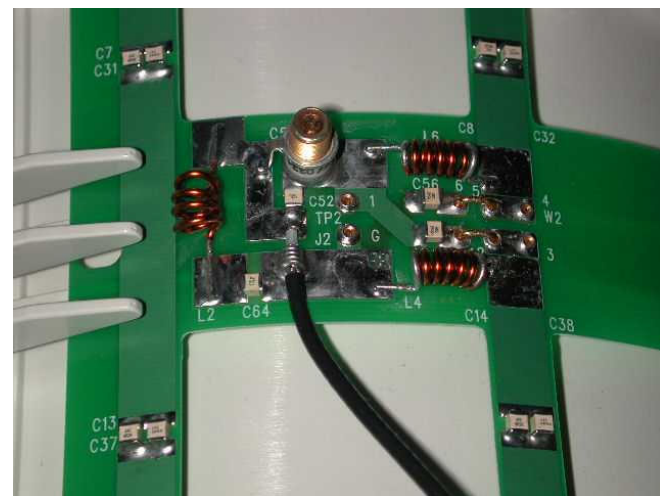
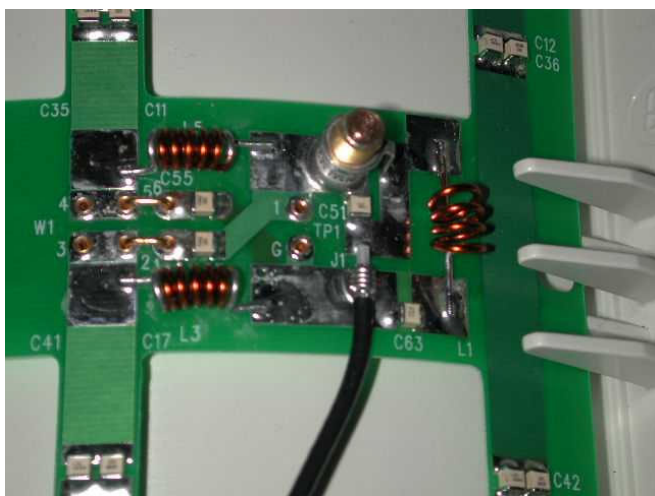
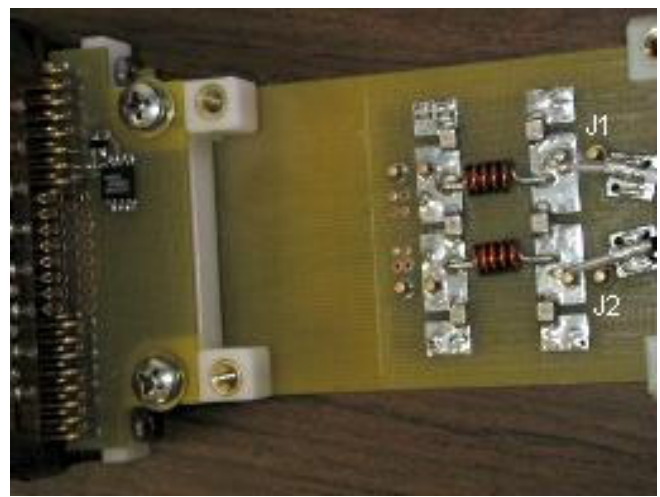
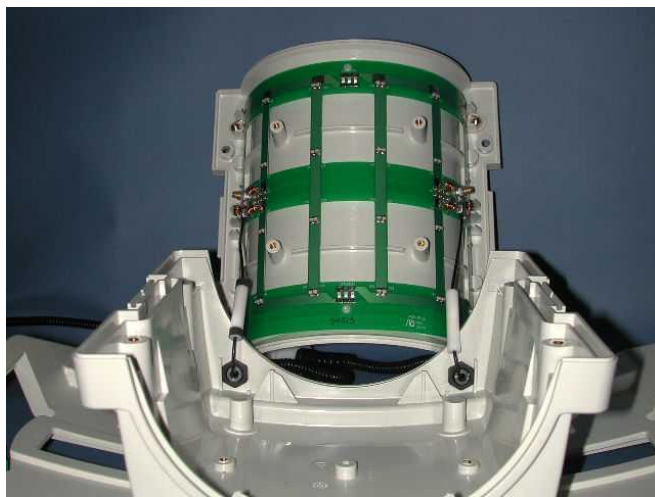
Note: Flex the cable while checking to test for intermittent opens or shorts in the cable.

- a) A continuity check for Receive should be measured from J2 center pin on the Interface Box PCB to the J1 center pin on the Bottom Coil PCB.
 - b) A continuity check for Receive should be measured from J2 Gnd on the Interface Box PCB to the J1 Gnd on the Bottom Coil PCB.
 - c) A continuity check for Transmit should be measured from J1 center pin on the Interface Box PCB to the J2 center pin on the Bottom Coil PCB.
 - d) A continuity check for Transmit should be measured from J1 Gnd on the Interface Box PCB to the J2 Gnd on the Bottom Coil PCB.
 - e) On the Interface Box PCB, verify open (No short circuit) using DMM between J1 center pin and J1 Gnd.
 - f) On the Interface Box PCB, verify open (No short circuit) using DMM between J2 center pin and J2 Gnd.
6. If the cable fails either test, remove the cable assembly per Section 5-2, External Cable Replacement. Affix the Bottom Cover of the Interface Box Assembly using four (4) screws and return for repair.
 7. If the cable passes the tests, affix the Bottom Cover of the Interface Box Assembly using four (4) screws.
 8. Carefully fit the Bottom Inner Assembly into the Bottom Cover.
 9. Install four (4) large nylon screws from the Bottom Coil Assembly.

10. Seat the Top Coil Assembly to the Bottom Coil Assembly.

EXTERNAL CABLE EXPECTED READINGS – TABLE 3-3

Signal	Positive Lead	Negative Lead	Reading
	J2 center pin on the Interface Box PCB	J1 center pin on the Bottom Coil PCB	0 Ohms
	J2 Gnd on the Interface Box PCB	J1 Gnd on the Bottom Coil PCB	0 Ohms
	J1 center pin on the Interface Box PCB	J2 center pin on the Bottom Coil PCB	0 Ohms
	J1 Gnd on the Interface Box PCB	J2 Gnd on the Bottom Coil PCB	0 Ohms
	J1 center pin on the Interface Box PCB	J1 Gnd on the Interface Box PCB	Open
	J2 center pin on the Interface Box PCB	J2 Gnd on the Interface Box PCB	Open



3-4 PIN Diodes Check

N/A

3-5 Mechanical Hardware Check

Lift the coil locks on both sides of the coil, and slide the coil from side to side. The coil must slide freely back and forth on the base with no binding or grinding. Lock the coil in the center of the base. Check the bottom of the base to ensure the 4 rubber feet are tight.

3-6 Troubleshooting Tips

The following is a list of common problems and solutions for those problems.

3-6-1 Receiving No Signal

Problem:

You are unable to pre-scan **or** are scanning and yet receiving no signal.

Possible Solution:

Verify that you have selected the appropriate system coil selection. Refer to Section 2-1-2 for additional information.

Verify the cable is correctly connected to the system and that it is completely plugged in.

Verify that the landmark is correct and that the cradle has not unlatched.

Verify that the scan locations and any FOV offsets are correct.

Perform a Continuity Check on the external cable (to be performed by a G.E. authorized Service Engineer only).

If you still cannot get a signal, try to scan (transmit and receive) with the body coil. For this test, be sure to remove the imaging coil from the magnet bore before you scan with the body coil. If you still receive no signal the problem probably lies with your MR system. If the body coil scan is satisfactory, attempt a scan using both another coil of the exact same type (receive-only, phased array or transmit/receive, whichever applies) and the same system coil selection. If the scan completes successfully, there is probably a problem with the Medical Advances coil. Contact IGC-Medical Advances Inc. for further assistance. If you are unable to scan with the substitute coil, there may be a system problem related to this particular coil type.

3-6-2 Image Quality

Problem:

The ratio obtained in the periodic quality assurance check is not greater than 85%, or the image quality is not what you expect it should be, given the parameters selected.

Possible Solution:

Review the selected protocol. If you are performing the Periodic Quality Assurance, be sure your protocol is **identical** to the protocol provided on Table 3-2-4 of this manual. If you are performing diagnostic images, you may need to increase NEX or FOV.

Perform a Continuity Check on the external cable (to be performed by a G.E. authorized Service Engineer only).

Verify that there are no loops in the cables.

Verify that there are no metal or ferromagnetic objects close to the coil, patient or magnet (i.e., safety pin, hair pin).

Verify that the coil is properly positioned.

Verify that your center frequency is within the frequency adjustment range for your system.

Verify that the R1, R2 and TG values from the pre-scan are within normally expected ranges.

Cleaning the electrical contacts that connect the upper and lower halves with Isopropyl Alcohol and a cotton swab has been demonstrated to improve the coil's SNR on occasion.

If you have not done so already, perform a system Quality Assurance phantom test, as outlined in Section 3 of this manual. If the values you obtain do not fall within normal operating parameters, investigate this further by performing a phantom scan with the body coil. **For this test, be sure to remove the imaging coil from the magnet bore before you scan with the body coil.** If you still have the same problems, there is probably an MR system problem. If the body coil scan is satisfactory, acquire a scan using **both** another coil of the **exact same type** (receive-only, phased array or transmit/receive, whichever applies) and the same system coil selection. If the image quality is visibly improved, there may be a problem with the Medical Advances coil. Contact IGC-Medical Advances Inc. for further assistance. If the image quality still suffers, there may be a system problem related to imaging with this type of coil.

3-6-3 Artifacts

Problem:

There is a black line or signal void on the image.

Possible Solution:

Verify that there is no metal present in the area being scanned in or on the patient.

Problem:

Some or all of the images appear shaded or exhibit uneven signal or banding.

Possible Solution:

Confirm that no metallic objects are located nearby, outside the FOV. This is especially important on images utilizing Fat Saturation.

If Fat Saturation is being used, verify that the CFA fine adjustment has been optimized.

For quadrature coils:

Verify that the coil is positioned with the cable exiting towards the bore and that the system and coil polarity match.

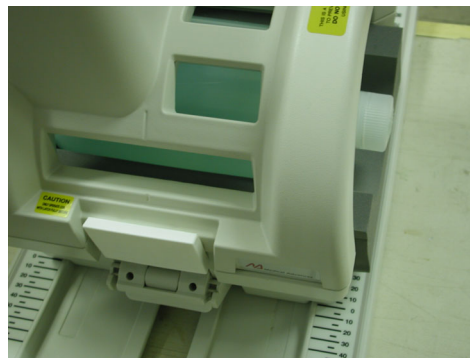


Figure 3-2-3-3: Coil alignment

SECTION 4 - MAINTENANCE

4-1 Coil Care

WARNING!

Detach coil connector from scanner before attempting to clean. Do not reattach after cleaning until coil has dried completely. Having the coil attached to the system during cleaning or when it is wet may result in electrical shock.

CAUTION

The Signa EXCITE 3.0T Quadknee Coil and patient comfort pads **must** be cleaned and stored using the following procedures:

Wipe with a cloth that has been dampened in a solution of 10% bleach and 90% tap water, or 30% isopropyl alcohol and 70% tap water.

Do not spray or pour any cleaning solution directly on the coil!

Let the coil housing and pads dry before use.

Store the coil in an air-conditioned scan room or equipment room.

Under no circumstances should the coil be placed into any type of sterilizer!

4-2 Special Care Requirements

N/A

SECTION 5 - REPLACEMENT

Simple removals that are clearly obvious are not described here.

Unless otherwise noted, the steps for re-assembly are simply the reverse order of the steps described for disassembly.

5-1 Disassembly of Coil

- 1) Open the Draw Latches and remove the Top Coil Assembly and set aside.
- 2) Remove the four large screws that hold the Bottom Inner Coil to the Bottom outer Housing.
- 3) Carefully lift the Bottom Inner Assembly out of the Bottom Cover and set it down on the Cable Assembly.

5-2 External Cable Replacement

Removal

- 1) Perform Section 5.1, Disassembly of Coil.
- 2) Unthread the Connector Cover Caps from the Connector Covers on both cables.
- 3) Separate the connector pins on both cables.
- 4) Carefully set the Bottom Inner to the side.
- 5) Remove the screw in Cable Nut A from both sides.
- 6) While holding the one cable with a 15mm flat, open end wrench, turn Cable Nut B counter clockwise using pliers.
- 7) Repeat for other cable.
- 8) Pull the cable assembly out of the Bottom Housing.

Reinstallation

Note: For Reverse coils the cable with the black Connector Cover Cap enters the mounting hole on the left.

- 1) Insert the new cable into the Bottom Housing ensuring the cable with the black Connector Cover Cap enters the mounting hole on the right.
- 2) Slide Cable Nut A onto the cable and ensure it seats in notch with the threaded hole up, then thread Cable Nut B onto the Cable, using a 15mm flat, open end wrench and pliers. Ensure that when tight, Cable Nut B's notch is centered over the threaded hole in Cable Nut A.
- 3) Repeat for the remaining cable.
- 4) Place the Bottom inner in a position so that the connector pins can be mated together.
- 5) Thread the Connector Cover Cap into the Connector Cover on both cables.
- 6) Carefully set the Bottom Inner into the Bottom Housing and secure with the four large screws that were removed.
- 7) Seat the Top Coil Assembly onto the Bottom Coil Assembly and secure with the Draw Latches.

5-3 Mechanical Hardware Replacement

See Section 6-1, Field Replaceable Units.

SECTION 6 - RENEWAL PARTS

6-1 Field Replaceable Units

FIELD REPLACEABLE UNITS LIST – TABLE 6-1-1

	Description	GE Part#	Supplier Part#
1	GE Signa® EXCITE™ 3.0T Foot/Knee Assembly w/ Cable	2380635-2	14280
2	Signa® EXCITE™ 3.0T Knee/Foot Cable Assembly	2380635-5	14281
3	Phantom Assembly	2357819-12	14246
4	Phantom Positioner (Not Shown)	2357819-11	14270
5	3T Extremity Coil Repair Kit (See Table 6-1-2)	2357819-10	14234

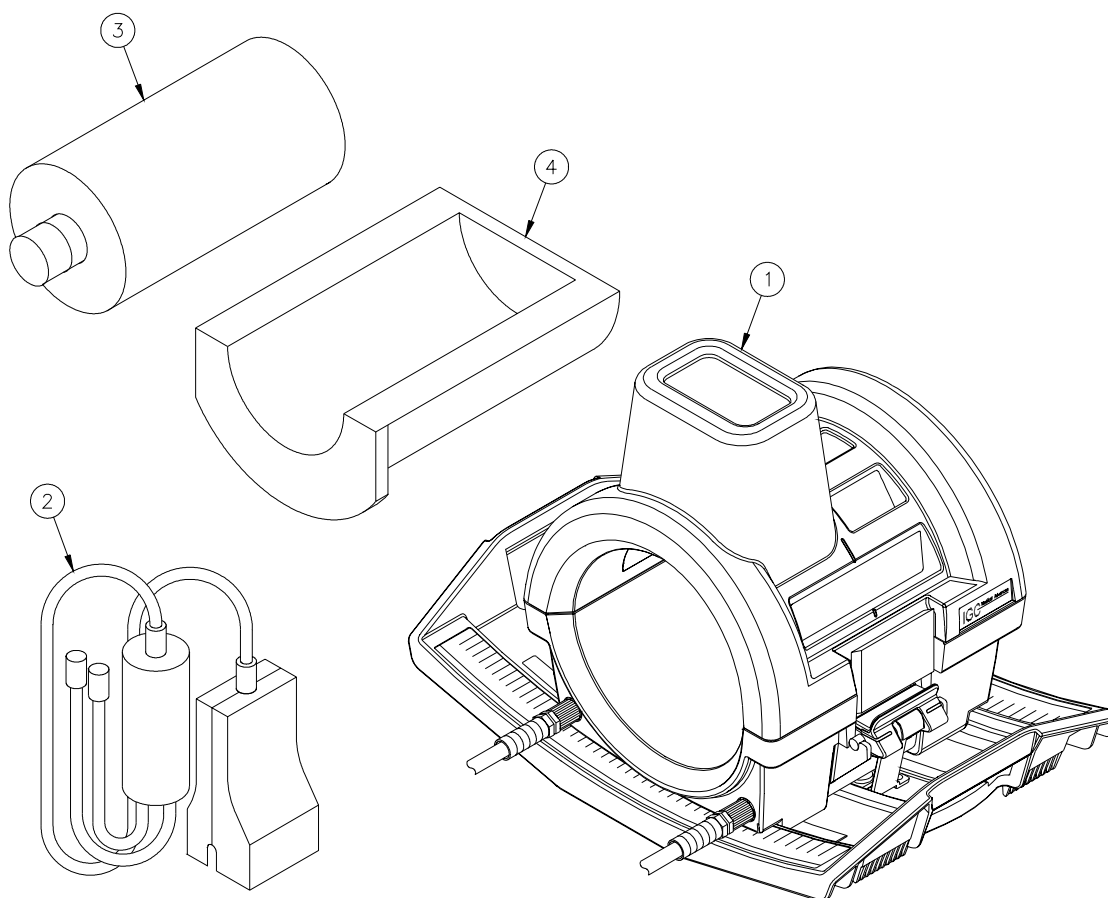


Figure 6-1-1: Field Replaceable Units

CONTENTS OF SPARE PARTS REPAIR KIT – TABLE 6-1-2

Item	QTY	Description	Supplier Part#
1	2	Draw Latch	20074P2
2	4	#6-32 x 0.438 FHMS, Brass	20117P25
3	2	Handle, Coil Lock	19711P1
4	2	Cover, Lock Handle	19712P1
5	4	#4-40 x 0.375 PHMS, Brass	20136P2
6	1	T-Bracket	19713P1
7	3	#10-32 x 0.375 PHMS, Brass	20136P14
8	4	Rubber Bumper	19727
9	4	#6-32 x 0.50 PHMS, Brass	20136P6

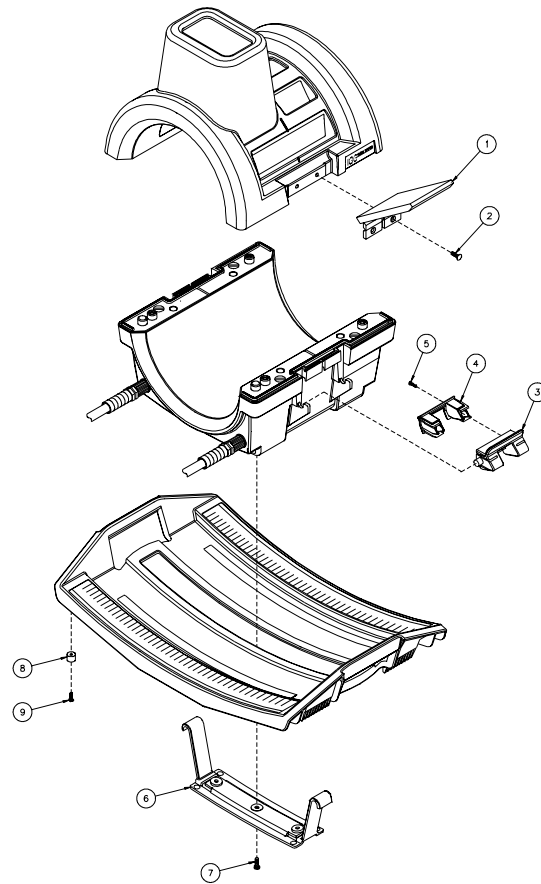


Figure 6-1-2: Spare Parts Repair Kit

6-2 Other Replaceable Accessories

TABLE 6-2: OTHER REPLACEABLE ACCESSORIES LIST

Description	GE Part#	Supplier Part#
Knee Pad	2357819-3	14629
Leg/Foot Support Pad	2357819-4	14627
Foot Support Pad	2357819-5	14626
Toe Pad	2357819-6	14628

SECTION 7 - APPENDIX

7-1 SNR Data Sheet

Use the table provided below to record the calculated signal to noise ratio (SNR) data obtained from the Functional Checks section.

Table 7-1: SNR Data Sheet

[illegible]

$$SNR = \frac{\text{Mean of Signal within ROI}}{\text{Std.Deviation of Noise within ROI}}$$

7-2 Schematic

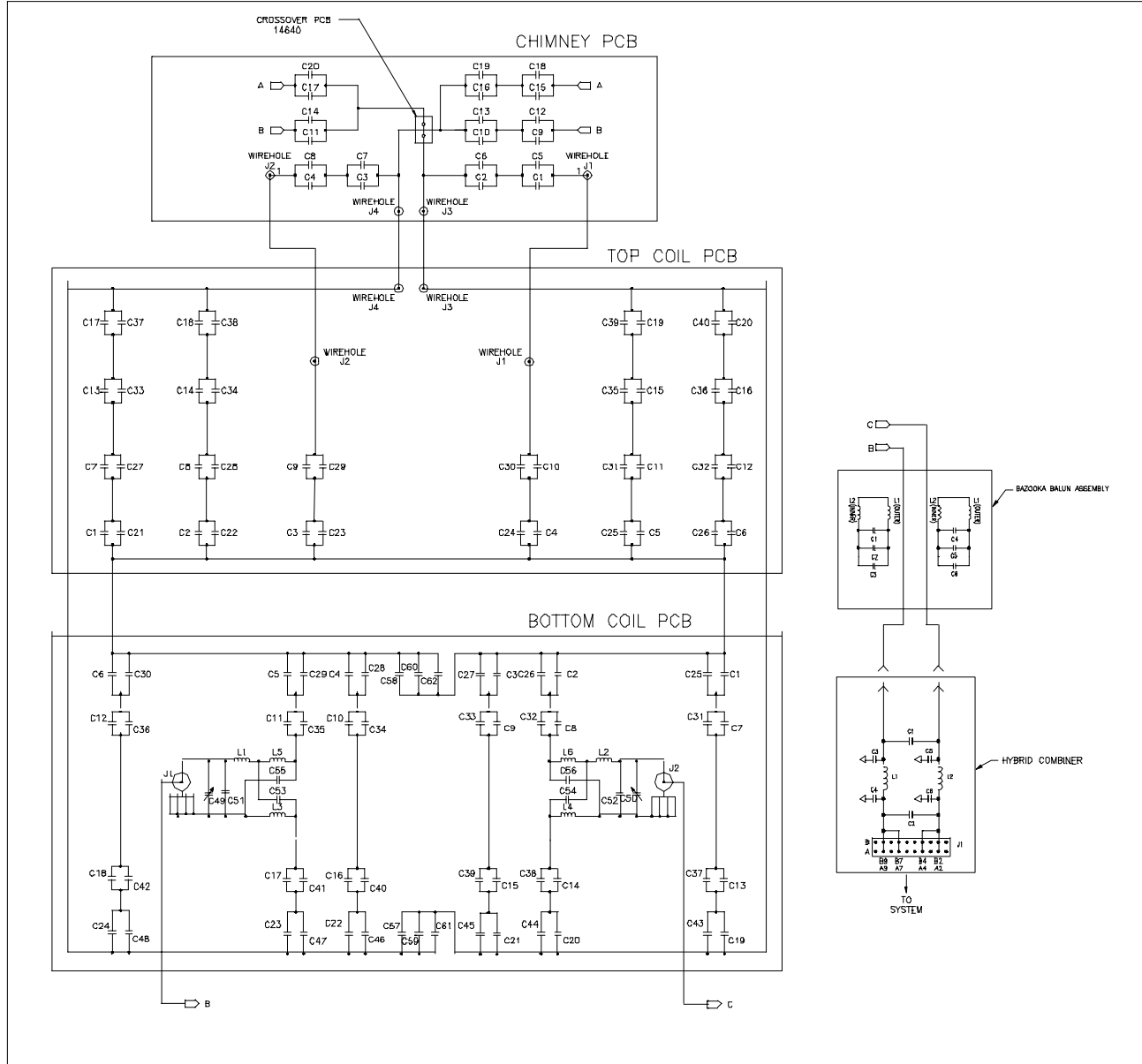


Figure 7-2: Signa EXCITE 3.0T Extremity Coil Schematic

7-3 Coil Configuration

Note: This coil configuration is only current as of the printing of this manual.

Parameter	Signa EXCITE 3.0T
coilName	QUADKNEE
coilCode	ekecd4o92l2ks6mlc7f3d9gkl2aoth6
physicalCoilDescription	Medical Advances T/R knee/foot coil

Parameter	Signa EXCITE 3.0T
coilType	1
multiCoil	no
xmitAtten	27
fastTGstartRG	12
fastTGstartTA	90
cableLoss	1.3
coilLoss	0.064
reconScale	1.00
autoshimRcvr	-1
headDefaultFreqDir	1
mCoilSeparationDir	0
maxFOV	480.0
assetThresh	10000
nuclide	1H
scic_a	0
scic_s	0
scic_c	0
mcSwitchSel	0
mcBiasEnable	0
mcVoltConfig0	0
mcVoltConfig1	0
mcOpenCktConfig0	0
mcOpenCktConfig1	0
mcOpenCktConfig2	0
mcOpenCktConfig3	0
mcShortCktConfig0	0
mcShortCktConfig1	0
mcShortCktConfig2	0
mcShortCktConfig3	0

Parameter	Signa EXCITE 3.0T
mcChannelSelect	0
startRec	0
endRec	0
numRec	0
userInterfaceGroup	18
prescanWeight0	1.0
prescanWeight1	1.0
prescanWeight2	1.0
prescanWeight3	1.0
prescanWeight4	1.0
prescanWeight5	1.0
prescanWeight6	1.0
prescanWeight7	1.0
aps1_mod	0
aps1_mod_PStloc	0
aps1_mod_PSRloc	0
aps1_mod_fov	200
aps1_plane	1
opthickPS_mod	5
coilPercentExposed	2
optxpos	5
extremity	yes
mcReconEnable	0
mcPortEnable	0
mcErrorEnable	0
startFastRec	4
numFastRec	1
endFastRec	4
linearQuad	1

REVISION HISTORY

Rev	Date	Author	Primary Reason for Change
A	7/3/03	K. Meurer	Engineering pre-release
0	9/26/03	K. Meurer	Original issue. Update coil configuration table.